

2. Write a C program to calculate x to the power y using recursion.

```
#include <stdio.h>
```

```
// Function Prototype
```

```
int power(int, int);
```

```
int main(void)
```

```
{
```

```
    int x, y;
```

```
    printf("Enter numbers: ");
```

```
    scanf("%d %d", &x, &y);
```

```
    // Function call and output
```

```
    printf("%d ^ %d = %d\n", x, y, power(x, y));
```

```
    return 0;
```

```
}
```

// Recursive function definition

int power(int x, int y)

```
{  
    if (y == 0)  
    {  
        return 1; // Any number to power 0 is 1  
    }  
    else if (y == 1)  
    {  
        return x; // Base case  
    }  
    else  
    {  
        // Recursive step:  $x * x^{(y-1)}$   
        return (x * power(x, y - 1));  
    }  
}
```